

1. Administrative & Study Identification

Official Study Title: A Phase II, Single-Arm Study of De-Escalation and Treatment-Free Remission in Patients with Chronic Myeloid Leukemia Treated with Nilotinib in First-Line Therapy Followed By a Second Attempt after Nilotinib and Asciminib Combination: DANTE **Study Brief Title:** A Study of Full Treatment-free Remission in Patients With Chronic Myeloid Leukemia **Short Title/Acronym:** DANTE **Protocol Number:** CAMN107AIT15 **NCT Number:** NCT03874858 **Posting ID:** 332181817887867375 **Sponsor/Company Name:** Novartis / Novartis Pharmaceuticals **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Investigational Product:** Nilotinib (AMN107 / Tasigna) and Asciminib (ABL001 / Scemblix) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objectives: 1. To assess the Full Treatment-Free Remission (FTFR) rate 96 weeks after the start of the consolidation period of the Treatment-Free Remission 1 (TFR1) stage. 2. To assess the percentage of patients in treatment-free remission 48 weeks after starting a second attempt at treatment-free remission during the TFR2 stage.

Secondary Objectives: To evaluate the percentage of patients who remain in sustained Deep Molecular Response (DMR) at the end of the consolidation period (week 48 TFR1 stage); to evaluate the percentage of patients remaining in DMR at week 48, 96, and 144 of the TFR1 stage; to evaluate the percentage of patients in FTFR at week 144; and to evaluate the overall eligibility rate for the TFR2 period following failure of a first attempt.

2.2 Background & Rationale Treatment-free remission (TFR) is considered a primary therapeutic goal for patients with Chronic Myeloid Leukemia (CML) who achieve a sustained deep molecular response (DMR). This study aims to shed light on a stepwise frontline de-escalation approach by assessing the effect of reduced-dose nilotinib (half the standard dose) for 12 months before full discontinuation. Furthermore, the study evaluates whether combining asciminib with nilotinib after a failed first TFR attempt can lead to higher and more durable TFR rates during a second attempt (TFR2).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **N\$:** Approximately 124 adult patients **Number of Sites:** 27 to 29 centers in Italy
Estimated Study Start: 21-Mar-2019 **Estimated Primary Completion:** 14-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a diagnosis of Chronic Myeloid Leukemia in chronic phase (CML-CP) according to World Health Organization criteria. 2. Treated with first-line nilotinib 300 mg twice daily (BID) for ≥ 3 years. 3. Achieved sustained deep molecular response (DMR), defined as a molecular response ≥ 4.0 logs (MR4.0; i.e., $BCR::ABL1/ABL1$ International Scale [IS] $\leq 0.01\%$) for ≥ 1 year. 4. Eastern Cooperative Oncology Group Performance Status (ECOG PS) of 0-2.

4.2 Exclusion Criteria 1. Previous Tyrosine Kinase Inhibitor (TKI) therapy other than nilotinib (except very brief standard exposures prior to starting frontline nilotinib as defined by protocol). 2. Loss of Major Molecular Response (MMR) or failure to achieve sustained DMR prior to enrollment. 3. Known atypical transcripts or $BCR::ABL1$ mutations conferring clinical resistance to nilotinib or asciminib.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **TFR1 Stage:** De-escalation with nilotinib (ATC Code: L01EA03) 300 mg once daily (half standard dose) for 48 weeks, followed by TFR (complete discontinuation) for 96 weeks. * **TFR2 Stage:** For patients failing TFR1, re-induction with asciminib 40 mg BID and nilotinib 300 mg BID for 96 weeks, followed by a second attempt at TFR (TFR2) for 48 weeks. **Route of Administration:** Oral **Duration of Treatment:** Up to 144 weeks for the TFR1 stage; up to an additional 144 weeks for the TFR2 stage (96 weeks re-induction + 48 weeks TFR).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for CML as per institutional guidelines. **Prohibited:** Other concurrent Tyrosine Kinase Inhibitors (TKIs) or systemic anti-leukemic therapies outside of the assigned study regimen.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, \$BCR::ABL1\$ PCR testing
Baseline	Day 0	Enrollment, Initiation of Nilotinib De-escalation (TFR1) or Re-induction (TFR2)
Interim	Weeks 1-12	Biweekly/monthly visits: Safety Labs, \$BCR::ABL1\$ PCR monitoring, Efficacy Assessment
Maintenance	Week 12 to EoS	Every 12 weeks: Routine clinical assessment, \$BCR::ABL1\$ PCR monitoring, TFR tracking
End of Study	Week 144	Final Vital Signs, Exit Interview, IP Accountability, Final \$BCR::ABL1\$ tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Percentage of patients in Full Treatment-Free Remission (FTFR) 96 weeks after the start of the consolidation period of the TFR1 stage:** FTFR is defined as patients with Major Molecular Response (MMR) or better (i.e. \$BCR::ABL1 \le 0.1\%\$ IS), including those who have totally discontinued treatment and those treated with half the standard dosage.
- 2. Percentage of patients in treatment-free remission 48 weeks after starting a second attempt at treatment-free remission during the TFR2 stage:** TFR is defined as patients with MMR or better (i.e. \$BCR::ABL1 \le 0.1\%\$ IS).

7.2 Secondary & Exploratory Endpoints

- * Percentage of patients who remain in sustained Deep Molecular Response (DMR) at the end of the consolidation period (week 48 TFR1 stage).
- * Percentage of patients who remain in DMR at the end of the consolidation period (week 48 of TFR1 stage), at 96 weeks, and at 144 weeks after the start of the consolidation period of TFR1 stage.
- * Percentage of patients in full treatment-free remission 144 weeks after the start of consolidation (end of the TFR1 stage).
- * Percentage of patients eligible for the TFR2 period (sustained DMR) among those entering re-induction with asciminib + nilotinib.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via routine clinical evaluations and patient reporting during biweekly, monthly, and Q12W visits. Special attention is given to TKI withdrawal syndrome during TFR. **Serious Adverse Events (SAEs):** Standard reporting timeline to the sponsor (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** Internal safety monitoring managed by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / All enrolled patients entering the respective TFR1 or TFR2 consolidation periods. **Sample Size Justification:** \$N \approx 124\$ appropriately powered to detect clinically meaningful rates of FTFR maintenance after nilotinib de-escalation. **Handling of Missing Data:** Patients who require re-initiation of nilotinib/TKIs due to loss of MMR, or those discontinued from the study for any reason, will be considered as treatment failures (Non-responder imputation).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits managed by Novartis Pharmaceuticals across the participating Italian centers. **Audit Trail:** eCRF locking, tracking of clinical pathway adherence, and continuous data clarification mechanisms established by the sponsor.

11. Study Identifiers & Governance

Primary ID: CAMN107AIT15 **Registry Number:** NCT03874858 **Posting ID:** 332181817887867375 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** DANTE Study **Official Regulatory Title:** A Phase II, Single-Arm Study of De-Escalation and Treatment-Free Remission in Patients with Chronic Myeloid Leukemia Treated with Nilotinib in First-Line Therapy Followed By a Second Attempt after Nilotinib and Asciminib Combination **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING)

12. Clinical Framework (CML Specific)

Primary Condition: Chronic Myeloid Leukemia (Preferred Name: Philadelphia chromosome positive chronic myelogenous leukemia) **Diagnosis Threshold:** Ph+ CML in sustained Deep Molecular Response (DMR; \$BCR::ABL1/ABL1 \le 0.01\%\$ IS) **Medical Methodology:** TKI De-escalation and Treatment-Free Remission (TFR) Stepwise Optimization **Optimization Target:** Maintenance of Major Molecular Response (MMR; \$BCR::ABL1 \le 0.1\%\$ IS) without standard-dose TKI therapy

13. Study Procedures & Methodology

Management Pathway: Structured algorithm for TKI dose reduction followed by complete discontinuation, and a predefined re-induction salvage pathway (Asciminib + Nilotinib) for TFR

failures. **Education Protocol (HCP):** Stringent molecular monitoring (qRT-PCR) training and threshold management for restarting therapy upon MMR loss. **Education Protocol (Patient):** Counseling on the concept of Treatment-Free Remission, strict adherence to PCR scheduling, and identification of TKI withdrawal symptoms. **Quality Audit Mechanism:** Centralized/Standardized laboratory evaluation for precise molecular response (\$BCR::ABL1\$ IS) measurements.

14. Observation & Follow-Up Schedule

Total Duration: Up to 144 weeks for TFR1, with an additional 144 weeks for patients progressing to TFR2. **Onsite Visit Cadence:** Biweekly/monthly until week 12, then every 12 weeks. **Remote Monitoring Frequency:** Standard sponsor-defined intervals. **Checklist Review Interval:** Every 12 weeks (Q12W) to coincide with molecular monitoring assessments.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Insert PI Name]	_____	[DD-MMM-YYYY]	Clinical Operations Lead	[Insert COL Name]	_____	[DD-MMM-YYYY]
Lead Statistician	[Insert Statistician Name]	_____	[DD-MMM-YYYY]				